aerospace innovation with the expanding frontiers of telecommunication and data science.

5.6 Developments and Pioneers of Fourth-generation Drone Models

The fourth generation of drones represents the zenith of unmanned aerial evolution, integrating artificial intelligence, machine learning, swarm coordination, advanced stealth, and digital networking into cohesive aerial ecosystems. This phase signifies the transition from manually supervised UAVs to autonomous, decision-making aerial entities capable of executing missions in highly dynamic, multi-domain environments. This section outlines the technological evolution that gave rise to fourth-generation drones and identifies the key pioneers and notable UAV models that have redefined global airpower, research, and commercial innovation.

5.6.1 Evolution of fourth-generation UAV development

The development of fourth-generation UAVs began in the early 2010s, driven by the need for smarter, faster, and more autonomous systems capable of operating in contested and data-driven environments. Building upon the endurance, stealth, and ISR (intelligence, surveillance, and reconnaissance) capabilities of third-generation drones, this new generation incorporated AI-based autonomy, real-time data processing, and network-centric warfare integration. Key technological shifts that defined this evolution include:

- *Integration of artificial intelligence (AI) and machine learning (ML)*: UAVs began using AI to independently interpret sensor data, identify targets, optimize routes, and make tactical decisions without continuous operator input.